

Experiment 3

Kinematics of projectile motion

3.1 Introduction

When you threw a ball into the air and observe how it soared up and then returned back down. Can you determine maximum height it reaches and predict where it will land?

In this experiment, we will toss a ball in the air with a known velocity and incline angle and determine its trajectory, from which we use this information and our knowledge of physics to predict where the ball will land when it hits the floor.

In our experiment, we consider two-dimensional projectile motion, for which air resistance is negligible.

3.2 Objectives

- Identify and explain the properties of a projectile, such as acceleration due to gravity, range, maximum height, and trajectory.
- Investigate the dependence of the maximum height on the initial velocity and the incline angle.
- Investigate the dependence of the range on the initial velocity and the incline angle

3.3 Theory

A projectile is an object that is thrown or projected into the air with an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration and air resistance. As shown in Figure 0., let the initial velocity vector be given by $\vec{v}_o$ at angle α_0 with x-axis. Then, neglecting the air resistance, the components of position, velocity and acceleration of the projectile motion at any point $p(x,y)$ on the trajectory are given by:

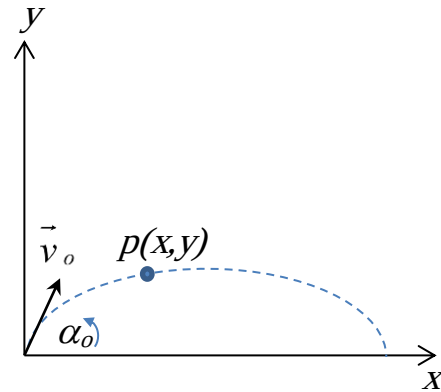


Figure 0.1: The path of projectile.

$$x = (v_o \cos \alpha_o) t \quad (3.1)$$

$$y = (v_o \sin \alpha_o) t - \frac{1}{2} g t^2 \quad (0.2)$$

$$v_x = v_o \cos \alpha_o \quad (0.3)$$

$$v_y = (v_o \sin \alpha_o) - g t \quad (0.4)$$

$$a_x = 0 \quad (0.5)$$

$$a_y = -g \quad (0.6)$$

Equations (0.5) and (0.6) show that we have substituted the acceleration due to gravity at the earth's surface, $-g$, for the vertical acceleration in the y direction and 0 for the horizontal acceleration in the x direction. At maximum height, the velocity v_y in equation

(0.4) is sitting to 0 and substituting the time into equation (0.2) we can get the maximum height of projection h as a function of the angle of projection α_0 as follows,

$$h = \frac{v_o^2 \sin^2 \alpha_o}{2g} \quad (0.7)$$

Similarly from equation (2.1) and (2.2) the maximum range x_m when $y = 0$ is given by,

$$x_m = \frac{v_o^2 \sin 2\alpha_o}{g} \quad (0.8)$$

3.4 Setup

The used setup is shown in figure 3.2.

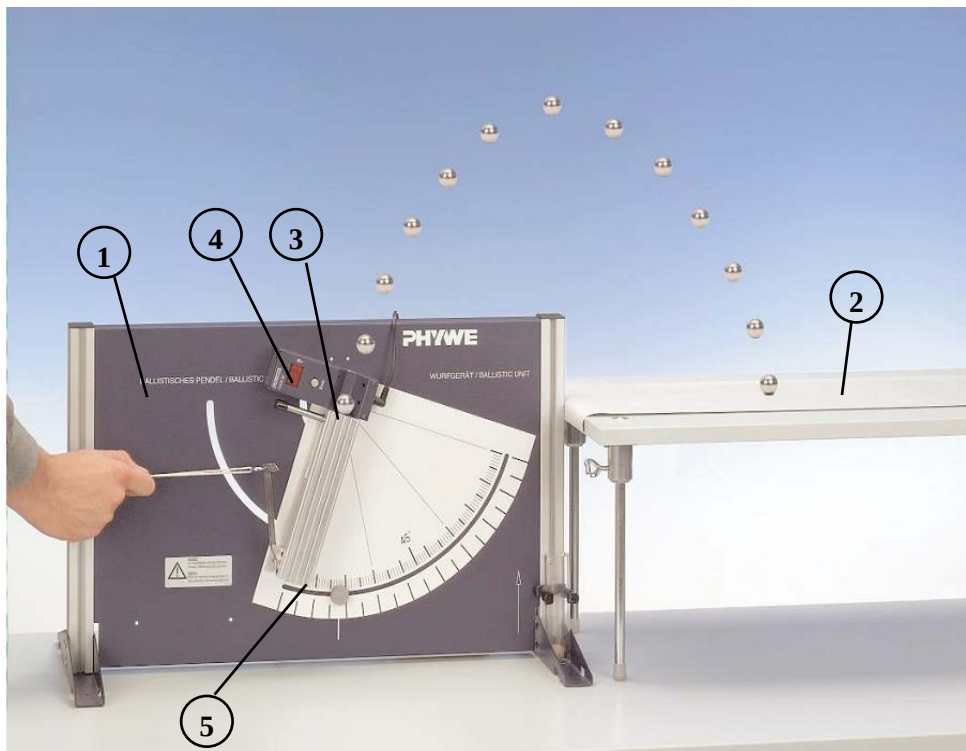


Figure 0.1: Projectile motion setup.

1 Ballistic unit

2 Recording paper, 1 roll, 25 m

4 Speed measuring attachment

5 firing spring "Catapult"

3 Steel ball, $d = 19 \text{ mm}$



Note that this Task is a demonstration only. Your TA will demonstrate the experiment and take the measurement to you. Please write the result down in your lab report.

3.5 Procedure

- Setup the ballistic unit as shown in Figure 0.1.
- To get a certain angle, turn the support base of the firing spring using the screw as shown in Figure 0. until you get the required angle. Start measurements with 30 degrees.

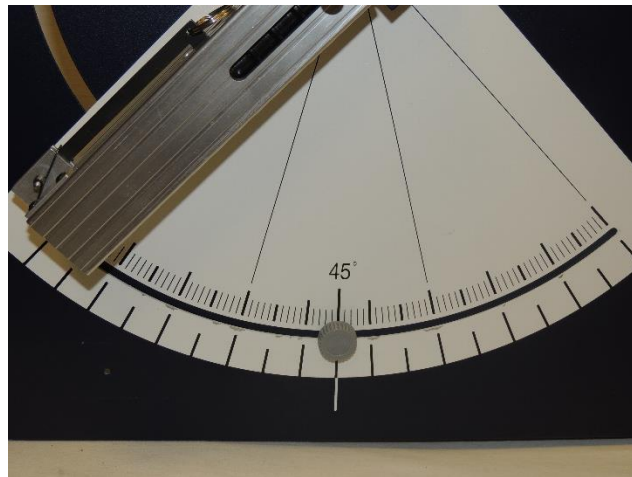


Figure 0.3: Adjusting the angle.

- Adjust 3 rulers at distances x_1 , x_2 and x_3 from the horizontal position of the firing spring as shown in Figure 0..

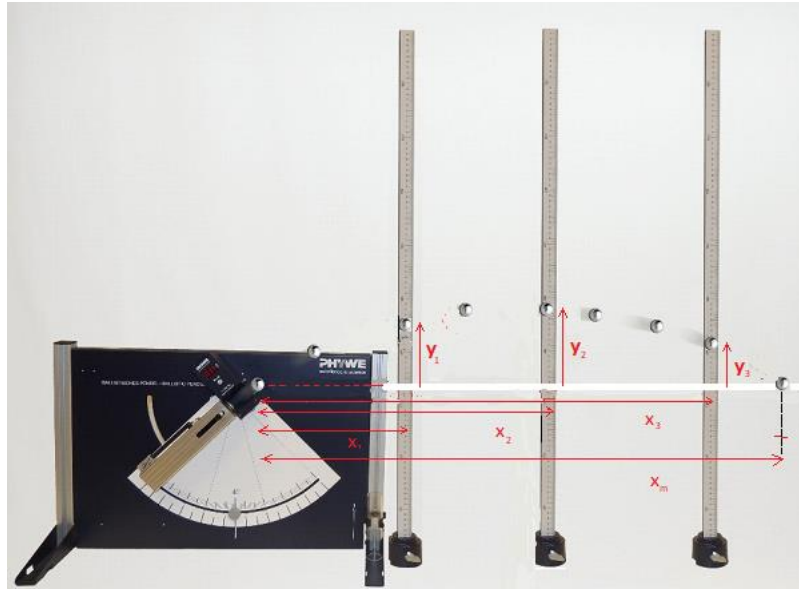


Figure 0.4: Adjusting the rulers.

- Adjust the initial velocity v_0 of the ball corresponding to the **three tension stages** of the firing spring that can be determined using the speed measuring attachment (4) .



Use firing protective glasses during firing the ball to protect your eyes.

- Release the firing spring .
- Measure the heights y_1 , y_2 and y_3 at which the ball crosses by the three rulers. You can use a camera with slow speed video to accurately determine the intersection points.
- Measure the distance between the firing spring and the point of impact on the table x_m “the range”.
- Record the data in the lab report.
- Repeat the steps from 2 to 7 at different angles $\alpha = 40, 50, 60, 70$ degrees.
- Plot graph between vertical and horizontal position of the ball at each angle.

- From the graph obtain, the maximum height, h , that can be attained at each angle.
- Plot graph (x_m, α) between range x_m and the corresponding angle ϕ .
- Plot graph (h, α) between maximum height and the corresponding angle α .
- Use Equation (0.7) to calculate the theoretical value of the maximum height h in terms of the projectile angle and the velocity measured by the speed tool. Record the values in the report and calculate the percentage error.
- Use Equation (0.8) to calculate the theoretical value of the range in terms of the projectile angle and the velocity measured by the speed tool. Record the values in the report and calculate the percentage error.
- Use the firing spring to change the velocity then repeat the task at the new velocity. Plot similar curves for the two different velocities on the same graph to see the effect of velocity in the maximum height and range.